

Power BI – Module End Evaluation

Q1: Data Cleaning and Loading

Imported the SalesData_1000Rows_WithIssues.csv dataset into Power BI using Home → Get Data → Text/CSV and opened it in Power Query Editor for further data cleaning and transformation.

Handling Missing Values:

Missing values were identified using Power Query's Column Quality and Column Statistics. Text fields such as CustomerName, Region, and Product were replaced with "Unknown". Missing numerical values were handled using the relationships between the fields rather than median imputation. Missing UnitPrice was calculated as $\text{Sales} \div \text{Quantity}$, missing Sales was calculated as $\text{Quantity} \times \text{UnitPrice}$, missing Cost was calculated as $\text{Sales} - \text{Profit}$, and missing Profit was calculated as $\text{Sales} - \text{Cost}$. The two missing OrderDate values were filled using the previous available date.

Duplicate Removal:

The dataset was checked for duplicate records across all columns. After selecting the complete dataset, duplicate rows were removed using Power Query's Remove Duplicates option. This reduced the dataset from 1,020 rows to 1,002 unique rows.

OrderDate Standardization:

The OrderDate column was converted to the Date data type in Power Query to ensure that all date values were stored consistently for further analysis. The two missing OrderDate values were filled using the date above as instructed.

Q2: Visualizations and Insights

Created a Pie Chart to show the distribution of orders across different regions. Used Region as the Legend and Count of OrderID as the Values. The chart displays the percentage contribution of each region to the total number of orders. unknown region was excluded from this visual.

A clustered column chart was created using Product on the X-axis and Count of OrderID on the Y-axis. A Top N filter was applied to display the 5 products with the highest number of orders, and the chart was sorted in descending order.

Created a Line Chart to show the profit trend over time. OrderDate was used as the time axis at the Year level, and Sum of Profit was used to display the total profit for each year.

Q3: DAX Calculations

Created a calculated table named `EastRegionOrders` using the DAX `FILTER` function to display only orders where the Region is East.

Created a calculated column named `ProfitCostDifference` using DAX to calculate the difference between Profit and Cost for each order using the formula: `ProfitCostDifference = Profit - Cost`.

Created a calculated measure named `TotalProfit` using the DAX `SUM` function to calculate the total profit earned from all orders.

Dashboard Development and Analysis

After completing the data cleaning, transformation, visualization, and DAX calculations, an interactive Power BI report was developed using multiple dashboard pages. The dashboards were designed to provide a clear and comprehensive view of sales, product, regional, and profitability performance. Consistent formatting, colour themes, slicers, navigation buttons, and interactive visualizations were used throughout the report.

1. Executive Sales Overview

The **Executive Sales Overview** dashboard was created to provide a high-level summary of overall business performance.

The dashboard includes KPI cards for:

- **Total Sales**
- **Total Profit**
- **Total Orders**
- **Total Quantity**
- **Profit Margin %**

The following visualizations were included:

- **Clustered Column Chart** – Sales vs Profit by Region
- **Area/Line Chart** – Profit Trend Over Time
- **Pie Chart** – Order Distribution by Region
- **Region Slicer**
- **Product Slicer**
- **OrderDate Slicer**

The dashboard also includes a **Key Insights** section highlighting important observations from the sales and profit data.

2. Product & Category Analysis

The **Product & Category Analysis** dashboard was developed to analyze product and category-level performance and identify products contributing significantly to sales, orders, profit, and quantity.

The dashboard includes:

- **Category → Product Treemap** showing sales distribution across categories and individual products.
- **Top 5 Trending Products by Orders** using a clustered column chart.
- **Top 5 Products by Profit** using a bar chart.
- **Top 5 Products by Quantity** using a column chart.
- **Category Slicer**
- **Product Slicer**

The Treemap provides a hierarchical view of **Category → Product**, allowing the user to analyze product performance within each category.

3. Regional Performance Analysis

The **Regional Performance Analysis** dashboard was created to compare sales, profit, and order distribution across different regions.

The dashboard contains:

- **Sales by Region** – Clustered Column Chart
- **Profit by Region** – Bar Chart
- **Order Distribution by Region** – Donut Chart
- **Monthly Sales Trend by Region** – Line Chart
- **Region Slicer**

This dashboard allows regional performance to be compared and helps identify differences in sales, profitability, order contribution, and sales trends across regions.

4. Profitability & Trend Analysis

The **Profitability & Trend Analysis** dashboard was developed to examine how sales, cost, and profit change over time.

The dashboard includes:

- **Profit Trend Over Time** – Line Chart
- **Sales Trend Over Time** – Area Chart
- **Sales vs Cost** – Clustered Column Chart

- **Profit Margin %** – Gauge
- **OrderDate Slicer**

An **OrderDate hierarchy** was used to provide drill-down analysis through:

Year → Quarter → Month

This allows the user to move from a yearly overview to quarterly and monthly performance.

Interactive Features and Navigation

The report was designed as an interactive multi-page Power BI dashboard. A navigation panel was added to allow users to move between the four dashboard pages.

The report provides interactive filtering through slicers for:

- Region
- Product
- Category
- OrderDate

Selections made in the slicers dynamically update the relevant visualizations, allowing users to explore specific portions of the dataset.

Drill-down functionality was also incorporated using the **OrderDate hierarchy**, enabling detailed analysis from yearly to quarterly and monthly levels. The **Category → Product** hierarchy in the Product & Category dashboard also allows product-level exploration within categories.

Key Insights

The completed dashboards provide several useful insights from the cleaned sales dataset:

- The **East region** has the highest share of orders among the identified regions.
- Regional sales values show differences in contribution across East, South, North, and West.
- The Product Analysis dashboard identifies the products with the highest number of orders.
- Product-level profit and quantity analysis highlights differences in product performance.
- Sales and profit show fluctuations across the analyzed time period.
- The Sales vs Cost analysis provides a comparison between revenue generated and associated costs.
- The Profit Margin KPI provides an overall measure of profitability.

